REVERSE LOGISTICS OPTIMIZATION: DESIGN OF A REVERSE SUPPLY CHAIN

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INTRODUCTION

- Rapid increase in waste due to shorter product life cycles
- Growing importance of sustainability in supply chain
- Reverse logistics enable recovery of value from waste
- This project focuses on optimizing reverse logistics
- networks

PURPOSE OF THE STUDY

- Designing an optimal reverse logistics network
- Minimizing collection, transportation, and processing costs
- Maximizing environmental sustainability
- Using MILP and MCDM (Multi-criteria Decision-making)

PROBLEM DEFINITION

- Increasing waste volumes create operational challanges
- Hight transportation and collection costs
- Inefficient routing and capacity utilization
- Incorrect facility location decisions

Core Problem: Designing a cost-effective and sustainable reverse logistics network

DECISION VARIABLES

- Selection of collection centers
- Activation of recycling facilities
- Capacity allocation decisions
- Material flow between facilities

METHODOLOGY

- Mixed-Integer Linear Programming (MILP)
- Cost minimization objective
- AHP for criteria weighting
- TOPSIS for ranking alternatives

LITERATURE REVIEW OVERVIEW

- Reverse logistics concepts
- Green supply chain management
- Network design approaches
- Optimization and routing studies

SCOPE AND LIMITATIOINS

- Single product type considered
- Linear transportation cost assumption
- Fixed facility capacities
- Regional-scale logistics network

EXPECTED BENEFITS

ECONOMIC BENEFITS

- Reduced transportation costs
- Improves facility utilization
- More afficient allocation of resources

ENVIRONMENTAL BENEFITS

- Lower carbon emissions through optimized routing
- Reduced environmental impact of waste

CONCLUSION

- Cost-effective reverse logistics network
- Sustainability-oriented optimization
- Decision support for facility planning
- Expandable for future studies

INFORMATION SOURCES

- Optimization of Reverse Logistics Network for End-of-Life Vehicles (Shangai, 2024)
- Waste Collection Vehicle Route Optimization – Hellenic Military Academy
- Reverse Logistics Network Design for E-Commerce Returns (2020)
- Reverse Logistics Optimization for Medical Waste (COVID19, 2023)
- Green Supply Chain Research – MDPI https://www.mdpi.com/journal/sustainability
- VRP and Routing Optimization – Wiley https://onlinelibrary.wiley.com/ doi/abs/10.1002/nav.3800080403
- Reverse Logistics Case Studies – Research Gate https://www.researchgate. net/

Atık Arazi Verileri

Düğüm (i)	Şehir	Atık Miktarı (Di)
1	İstanbul (Hadımköy)	100.6 Ton
2	Kocaeli (Şekerpınar)	71.9 Ton
3	Bursa	43.1 Ton
4	İzmir	28.8 Ton
5	Ankara	28.8 Ton
6	Eskişehir (Tepebaşı)	14.3 Ton
TOPLAM		287.5 Ton

Atık Miktarı (Di)	Gerekçe / Varsayım
100.6 Ton	Ana Merkez (%35 Ağırlık)
71.9 Ton	Sanayi Bölgesi (%25 Ağırlık)
43.1 Ton	Otomotiv Üssü (%15 Ağırlık)
28.8 Ton	Ege Bölge Dağıtım (%10 Ağırlık)
28.8 Ton	İç Anadolu Hub (%10 Ağırlık)
14.3 Ton	Yeşil Lojistik Deposu (%5 Ağırlık)
287.5 Ton	

Aday Tesis ve Kapasite Verileri

Tesis (j)	Şehir	Sabit Kurulum Maliyeti (Fj)
F1	İstanbul (Hadımköy)	**$50,000**
F2	Kocaeli (Şekerpınar)	**$40,000**
F3	Eskişehir (Tepebaşı)	**$25,000**
F4	Ankara	**$30,000**

Kapasite (Capj)	Notlar
200 Ton/Yıl	En yüksek kira ve işçilik maliyeti.
150 Ton/Yıl	
100 Ton/Yıl	Düşük maliyet + Teşvik avantajı.
120 Ton/Yıl	

Mesafe Matrisi

Kaynak \ Hedef	İst (Hadımköy)	Kocaeli
İstanbul	10 km	85 km
Kocaeli	85 km	10 km
Bursa	150 km	130 km
İzmir	480 km	450 km
Ankara	450 km	350 km
Eskişehir	280 km	200 km

Eskişehir	Ankara
280 km	450 km
200 km	350 km
150 km	380 km
410 km	590 km
235 km	10 km
10 km	235 km

Maliyet ve Emisyon Parametreleri

Parametre	Değer
Birim Taşıma Maliyeti (C_{var})	0.12
Emisyon Faktörü (E_{factor})	0.062
Karbon Vergisi / Maliyeti	50
Geri Kazanım Geliri	250

Birim	Kaynak / Hesaplama
USD / Ton-km	Sektör ortalaması (Mazot + Şoför).
kg CO_2 / Ton-km	Mars Logistics filo verileri ve Euro 6 motor standartları.
USD / Ton CO_2	AB Yeşil Mutabakatı (ETS) tahmini fiyatı.
USD / Ton	Atığın geri dönüştürülmesiyle elde edilen hurda değeri (Plastik/Metal).

THANK YOU FOR LISTENING TO US!